FloodGuard AI

Flood preparedness linked to community health and response logistics

DSH Hacks V2 · AI x Healthcare
Sylhet & Sunamganj · Bangladesh

01 · THE CHALLENGE

Flash floods in north-eastern Bangladesh can isolate villages, shelters and health facilities. Contaminated water, displacement and blocked clinic routes create urgent public-health and logistics needs alongside rising water.

FloodGuard AI brings hydrological context and healthcare preparedness into one bilingual decision-support interface.

02 · WHAT IT SUPPORTS

- Upstream Open-Meteo precipitation and a daily GloFAS modelled discharge estimate.
- Rule-based flood-risk and hand-weighted health indicators with 48-hour trend extrapolation.
- Illustrative shelter supply planning and modeled clinic-access routes.
- Community reports with consent, privacy protections and admin moderation.

03 · ADVANCED RESPONSE WORKFLOWS

- **Citizen check-ins:** safe / need-help / evacuating status with delete-my-data controls.
- **Alert subscriptions:** severity preferences and optional SMS outbox registration.
- **Responder desk:** help-request queue, acknowledge/resolve actions, SitRep archive and audit log.
- **Monitoring engine:** scheduled SQLite readings, rule cooldown/escalation and live server-sent events.
- **Forecasting:** OLS/Holt trend ensemble with an illustrative range band from recorded history (not a calibrated probability forecast).
- **Admin:** PIN-protected moderation, bilingual alert publishing and persistent warning/danger thresholds.

04 · DATA PROVENANCE

LIVE API Hourly precipitation from Open-Meteo. GloFAS is a daily ~5 km grid-cell model estimate, **not** a local gauge reading; the closest cell may not represent the intended river.

DERIVED Risk heuristics, OLS/ensemble forecasts, allocation and travel-lag calculations.

SIMULATED Station levels, shelter stocks and some clinic-road inputs. Demo history is labelled.

05 · TECHNOLOGY & AI

Stack: React 19, TypeScript, Vite, Tailwind CSS, Express, SQLite, Leaflet and Recharts.

Gemini: optional natural-language chat and prediction explanations. Safety-critical ORS and water-treatment instructions are fixed server-side content, not generated.

Language: English and Bangla interface support; PWA shell caching for weak connections.

06 · LIMITATIONS AND RESPONSIBLE USE

FloodGuard AI is a research/demo decision-support prototype, not an official warning service or medical device. Its health index is hand-weighted and not clinically validated. Model estimates and simulated values must not be presented as live gauge measurements. SMS stays queued unless an external gateway adapter is configured. Official authorities and licensed clinicians remain the source of operational instructions.

Emergency contacts shown in the app: 999 (national emergency), 1090 (disaster helpline), 16263 (health helpline). Confirm current contacts with official agencies before public deployment.

FloodGuard AI · Project summary · Updated 11 November 2026
AI-assisted development tools were used; generated code and claims require maintainer review.

Sources: Open-Meteo Flood API documentation · Node.js SQLite documentation · Bangladesh DGHS / DDM official service pages.